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# 337455(37)

## B. E. (Fourth Semester) Examination, Nov.-Dec. 2015

(New Scheme)

(Mech. Engg. Branch)

## NUMERICAL ANALYSIS & COMPUTER PROGRAMMING (C & C++)

*Time Allowed : Three hours*

*Maximum Marks : 80*

*Minimum Pass Marks : 28*

*Note : Attempt all questions. Part (a) from each question is compulsory. Attempt any two parts from parts (b), (c) and (d) of each question.*

### Unit-I

1. (a) Round off the number 8.8975 upto three decimal places. 2



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(b) Apply Gauss elimination method to solve the equation : 7

$$x+4y-z=-5$$

$$x+y-6z=-12$$

$$3x-y-z=4$$

(c) Find a root of the equation $x^3-4x-9=0$, using the bisection method in four stages. 7

(d) Using Newton's iterative method. Find the real root of $x\log_{10}x=1\cdot2$ correct to five decimal places. 7

### Unit-II

2. (a) Reduce the pattern $y=ae^{bx}$, where $a$ and $b$ are constants, into a linear law of the form $y=mx+c$. 2

(b) From the following table, estimate the number of students who obtained marks between 40 and 45 : 7

| Marks | No. of Students |
|---|---|
| 30-40 | 31 |
| 40-50 | 42 |



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| | |
|---|---|
| 50-60 | 51 |
| 60-70 | 35 |
| 70-80 | 31 |

(c) Interpolate by means of Gauss's backward formula, the population of a town for the year 1974, given that : 7

| Year | Population (in thousands) |
|---|---|
| 1939 | 12 |
| 1949 | 15 |
| 1959 | 20 |
| 1969 | 27 |
| 1979 | 39 |
| 1989 | 52 |

(d) Fit a second degree parabola to the following data : 7

| $x$ | $y$ |
|---|---|
| 0 | 1 |
| 1 | 1·8 |
| 2 | 1·3 |
| 3 | 2·5 |
| 4 | 6·3 |



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## Unit-III

3. (a) Write the Simpson's 3/8 rule. 2

(b) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using : 7

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule

(iii) Simpson's 3/8 rule

(c) Using Runge-Kutta method of fourth order, solve

$\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0 \cdot 2, 0 \cdot 4$. 7

(d) Using modified Euler's method, find an approximate value of $y$ when $x = 0 \cdot 3$, given that $\frac{dy}{dx} = x + y$ and $y = 1$ when $x = 0$. 7



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## Unit-IV

4. (a) Classify the following equation as elliptic, parabolic and hyperbolic : 2

$$\frac{\partial^2 u}{\partial x^2}+4\frac{\partial^2 u}{\partial x\,\partial y}+4\frac{\partial^2 u}{\partial y^2}-\frac{\partial u}{\partial x}+2\frac{\partial u}{\partial y}=0$$

(b) Given the value of $u(x, y)$ on the boundary of the square in the fig. (a) evaluate the function $u(x, y)$ satisfying the Laplace equation $\nabla^2 u = 0$ at the pivotal points of this fig. 7

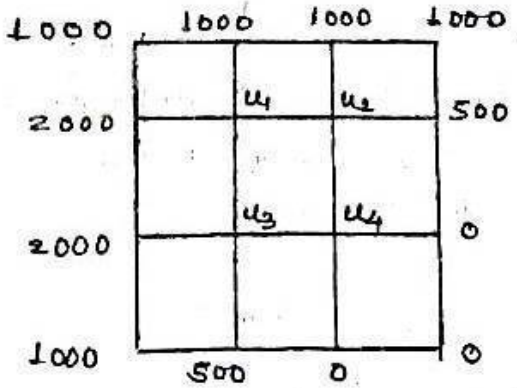


Fig. (a)



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(c) Find the values of $u(x,t)$ satisfying the parabolic equation $\frac{\partial u}{\partial t}=4\frac{\partial^2 u}{\partial x^2}$ and the boundary conditions $u(0,t)=0=u(8,t)$ and $u(x,0)=4x-\frac{1}{2}x^2$ at the points $x=i,\ i=0,1,2,\ldots\ldots\ldots\ldots,8$ and $t=\frac{1}{8}j : j=0,1,2,\ldots\ldots\ldots\ldots,5$. 7

(d) Evaluate the pivotal values of the equation $U_{tt} = 16\,U_{xx}$ taking $h \equiv 1$ upto $t = 1{\cdot}25$. The boundary conditions are $u(0, t) = u(5, t)$, $u_t(x, 0) = 0$ and $u(x, 0) = x^2(5 - x)$. 7

## Unit-V

5. (a) Define C-standard libraries. 2

(b) Write a program to find a number is odd and even. 7



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(c) Explain logical operators. 7

(d) What is array? How does it differ from ordinary variable. 7

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